Experiment : 5

❖ **Aim :**

To design and develop a Simple Blog Web Application using Flask that allows users to create, read, update, and delete blog posts.

❖ **Tasks :**

• **Project Setup & Introduction** – Create a project folder named simple_blog, install Flask, and create required files such as app.py, templates, and static folder.

• **Flask App Initialization** – Import Flask modules, initialize the application, and create a home route to display blog posts using a list or dictionary.

• **Create Blog Posts** – Create a form to add new blog posts with title and content, and store them in the application.

• **Read & Display Blog Posts** – Display all blog posts on the homepage with proper formatting and navigation links.

• **Update Blog Posts** – Add an edit option to update existing blog posts with pre-filled data.

• **Delete Blog Posts** – Provide a delete option to remove blog posts and update the list dynamically.

• **Basic Styling & Navigation (Bonus)** – Apply CSS for layout, fonts, and spacing, and add navigation links like Home and Create Post.

❖ **Tech Stack Used :**

• Python • Flask • HTML • CSS

❖ Html Code :

>Index.html :

```
{% extends "base.html" %}

{% block title %}Home — Simple Blog{% endblock %}

{% block content %}

<h1 class="page-heading">Latest Posts</h1>

{% if posts %}
```

```
<!-- Loop over every post and render a card -->
{% for post in posts|reverse %}
<article class="post-card">

<h2>{{ post.title }}</h2>
<p class="post-meta">Published on {{ post.date }}</p>
<p class="post-content">{{ post.content }}</p>

<div class="post-actions">
<!-- Edit button: navigates to the edit form -->
<a href="{{ url_for('edit', post_id=post.id) }}" class="btn btn-secondary">
✏️ Edit
</a>

<!-- Delete button: submits a POST form to delete route -->
<form action="{{ url_for('delete', post_id=post.id) }}" method="POST"
onsubmit="return confirm('Delete this post? This cannot be undone.');">
<button type="submit" class="btn btn-danger">🗑️ Delete</button>
</form>
</div>

</article>
{% endfor %}

{% else %}
<!-- Empty state when no posts exist -->
<div class="empty-state">
<div class="empty-icon">📝</div>
<p>No posts yet. Be the first to write something!</p>
<a href="{{ url_for('create') }}" class="btn btn-primary">Create Your First
Post</a>
</div>
{% endif %}

{% endblock %}
```

>**Base.html** :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
<title>{% block title %}Simple Blog{% endblock %}</title>

<!-- Google Fonts: Playfair Display + DM Sans -->
<link rel="preconnect" href="https://fonts.googleapis.com"/>
<link
href="https://fonts.googleapis.com/css2?family=Playfair+Display:ital,wght@0,7
00;1,400&family=DM+Sans:wght@300;400;500&display=swap"
rel="stylesheet"/>

<style>
/* — CSS Reset & Variables ———————————————— */
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }

:root {
--ink:      #1a1612;
--paper:    #f5f0e8;
--cream:    #ede7d8;
--rust:     #b84a2e;
--rust-lt:  #d4603f;
--muted:    #7a7060;
--border:   #cec4b0;
--shadow:   rgba(26,22,18,.12);
}

html { scroll-behavior: smooth; }

body {
font-family: 'DM Sans', sans-serif;
background: var(--paper);
color: var(--ink);
min-height: 100vh;
display: flex;
flex-direction: column;
}

/* — Masthead / Header ———————————————————— */
```

```
header {
background: var(--ink);
color: var(--paper);
padding: 0 2rem;
border-bottom: 3px solid var(--rust);
}

.header-inner {
max-width: 900px;
margin: 0 auto;
display: flex;
align-items: center;
justify-content: space-between;
padding: 1.4rem 0;
gap: 1rem;
}

.logo {
font-family: 'Playfair Display', serif;
font-size: 1.7rem;
letter-spacing: .02em;
color: var(--paper);
text-decoration: none;
}

.logo span { color: var(--rust-lt); font-style: italic; }

nav { display: flex; gap: .5rem; }

nav a {
font-size: .85rem;
font-weight: 500;
text-transform: uppercase;
letter-spacing: .08em;
color: var(--cream);
text-decoration: none;
padding: .45rem .9rem;
border: 1px solid transparent;
border-radius: 3px;
transition: background .2s, border-color .2s, color .2s;
}

nav a:hover,
nav a.active {
```

```
background: var(--rust);
border-color: var(--rust);
color: #fff;
}

/* — Main Content ———————————————————————————— */
main {
flex: 1;
max-width: 900px;
width: 100%;
margin: 0 auto;
padding: 2.5rem 2rem;
}

/* — Page heading ———————————————————————————— */
.page-heading {
font-family: 'Playfair Display', serif;
font-size: 2rem;
margin-bottom: 1.8rem;
padding-bottom: .8rem;
border-bottom: 2px solid var(--border);
color: var(--ink);
}

/* — Flash / Error banner ———————————————————— */
.alert {
background: #fdf0ee;
border-left: 4px solid var(--rust);
color: var(--rust);
padding: .75rem 1rem;
border-radius: 0 4px 4px 0;
margin-bottom: 1.4rem;
font-size: .9rem;
}

/* — Cards ———————————————————————————————— */
.post-card {
background: #fff;
border: 1px solid var(--border);
border-radius: 6px;
padding: 1.6rem 1.8rem;
margin-bottom: 1.4rem;
box-shadow: 0 2px 8px var(--shadow);
transition: transform .15s, box-shadow .15s;
```

```
}

.post-card:hover {
transform: translateY(-2px);
box-shadow: 0 6px 20px var(--shadow);
}

.post-card h2 {
font-family: 'Playfair Display', serif;
font-size: 1.35rem;
margin-bottom: .4rem;
}

.post-meta {
font-size: .78rem;
color: var(--muted);
text-transform: uppercase;
letter-spacing: .07em;
margin-bottom: .85rem;
}

.post-content {
line-height: 1.75;
color: #3a342c;
font-size: .95rem;
}

.post-actions {
display: flex;
gap: .6rem;
margin-top: 1.2rem;
padding-top: 1rem;
border-top: 1px solid var(--cream);
}

/* — Buttons ———————————————————————————————— */
.btn {
display: inline-block;
font-family: 'DM Sans', sans-serif;
font-size: .82rem;
font-weight: 500;
text-transform: uppercase;
letter-spacing: .07em;
padding: .48rem 1.1rem;
```

```
border-radius: 3px;
border: none;
cursor: pointer;
text-decoration: none;
transition: background .2s, color .2s;
}

.btn-primary {
background: var(--rust);
color: #fff;
}
.btn-primary:hover { background: var(--rust-lt); }

.btn-secondary {
background: var(--cream);
color: var(--ink);
border: 1px solid var(--border);
}
.btn-secondary:hover { background: var(--border); }

.btn-danger {
background: transparent;
color: #b03a22;
border: 1px solid #e0b0a8;
}
.btn-danger:hover { background: #fdf0ee; }

/* — Forms ———————————————————————————————— */
.form-card {
background: #fff;
border: 1px solid var(--border);
border-radius: 6px;
padding: 2rem 2.2rem;
box-shadow: 0 2px 8px var(--shadow);
max-width: 680px;
}

.form-group { margin-bottom: 1.4rem; }

label {
display: block;
font-size: .82rem;
font-weight: 500;
text-transform: uppercase;
```

```
letter-spacing: .07em;
color: var(--muted);
margin-bottom: .5rem;
}

input[type="text"],
textarea {
width: 100%;
padding: .7rem .9rem;
font-family: 'DM Sans', sans-serif;
font-size: .95rem;
color: var(--ink);
background: var(--paper);
border: 1px solid var(--border);
border-radius: 4px;
outline: none;
transition: border-color .2s, box-shadow .2s;
resize: vertical;
}

input[type="text"]:focus,
textarea:focus {
border-color: var(--rust);
box-shadow: 0 0 0 3px rgba(184,74,46,.1);
}

textarea { min-height: 160px; }

.form-actions { display: flex; gap: .7rem; margin-top: 1.6rem; }

/* — Empty state ———————————————————————————— */
.empty-state {
text-align: center;
padding: 4rem 2rem;
color: var(--muted);
}

.empty-state .empty-icon { font-size: 3rem; margin-bottom: 1rem; }
.empty-state p { margin-bottom: 1.5rem; line-height: 1.6; }

/* — Footer ———————————————————————————————— */
footer {
background: var(--ink);
color: var(--muted);
```

```
text-align: center;
padding: 1.1rem;
font-size: .8rem;
letter-spacing: .05em;
}
</style>
</head>
<body>

<!-- — Header / Navigation ———————————————— -->
<header>
<div class="header-inner">
<a href="{{ url_for('index') }}" class="logo">The <span>Inkwell</span></a>
<nav>
<a href="{{ url_for('index') }}"
class="{% if request.endpoint == 'index' %}active{% endif %}">
Home
</a>
<a href="{{ url_for('create') }}"
class="{% if request.endpoint == 'create' %}active{% endif %}">
+ New Post
</a>
</nav>
</div>
</header>

<!-- — Page Content ———————————————————— -->
<main>
{% block content %}{% endblock %}
</main>

<!-- — Footer ———————————————————————— -->
<footer>
Simple Blog &mdash; Experiment-5 &mdash; Built with Flask
</footer>

</body>
</html>
```

>Create.html :

```
{% extends "base.html" %}

{% block title %}Create Post — Simple Blog{% endblock %}

{% block content %}

<h1 class="page-heading">Create New Post</h1>

<div class="form-card">

<!-- Show validation error if present -->
{% if error %}
<div class="alert">⚠ {{ error }}</div>
{% endif %}

<!--
POST to /create
Fields: title, content
-->
<form action="{{ url_for('create') }}" method="POST">

<div class="form-group">
<label for="title">Post Title</label>
<input
type="text"
id="title"
name="title"
placeholder="Enter a compelling title..."
value="{{ request.form.get('title', '') }}"
required
/>
</div>

<div class="form-group">
<label for="content">Content</label>
<textarea
id="content"
name="content"
placeholder="Write your post here..."
required
```

```
>{{ request.form.get('content', '') }}</textarea>
</div>

<div class="form-actions">
<button type="submit" class="btn btn-primary">Publish Post</button>
<a href="{{ url_for('index') }}" class="btn btn-secondary">Cancel</a>
</div>

</form>
</div>

{% endblock %}
```

CSS Code :

```
/*
static/style.css
Simple Blog — Experiment-5
Additional / override styles (base styles live in base.html)
*/

/* Responsive: narrow screens */
@media (max-width: 600px) {
.header-inner {
flex-direction: column;
align-items: flex-start;
gap: .6rem;
}

nav { flex-wrap: wrap; }

main { padding: 1.5rem 1rem; }

.form-card { padding: 1.4rem 1.2rem; }

.post-card { padding: 1.2rem 1rem; }
}
```

❖ App.py Code :

```
from flask import Flask, render_template, request, redirect, url_for

from datetime import datetime


# --- App Initialization ---

app = Flask(__name__)


# --- In-Memory Storage ---
```

```
# Each post is a dict: { id, title, content, date }
posts = [
{
"id": 1,
"title": "Welcome to Simple Blog",
"content": "This is your first blog post! Use the navigation above to create, edit, or delete posts. This platform was built with Flask as part of Experiment-5.",
"date": "April 2, 2026"
},
{
"id": 2,
"title": "Getting Started with Flask",
"content": "Flask is a lightweight Python web framework. It's perfect for building small to medium web applications quickly and cleanly. Combined with Jinja2 templates, it makes web development a breeze.",
"date": "April 2, 2026"
}
]

# Counter to generate unique post IDs
next_id = 3



# -------------------------------------------------------
# HOME ROUTE — Read: Display all posts
# -------------------------------------------------------
@app.route("/")
```

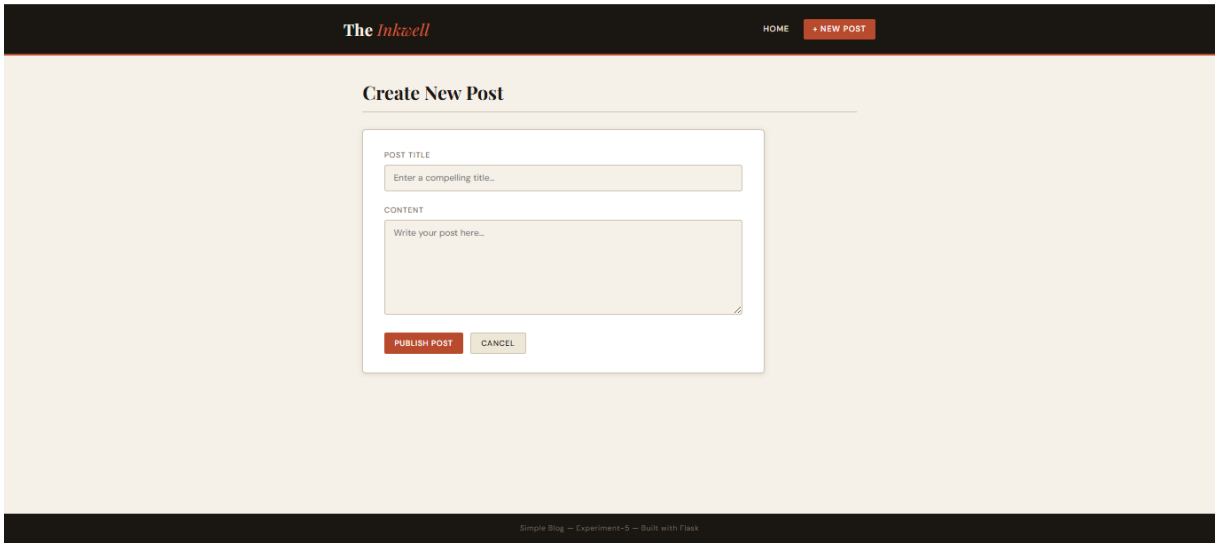
```
def index():
"""Render the home page with all blog posts."""
return render_template("index.html", posts=posts)



# -------------------------------------------------------
# CREATE ROUTE — Add a new blog post
# -------------------------------------------------------
@app.route("/create", methods=["GET", "POST"])
def create():
"""
GET  -> Show the create post form.
POST -> Validate input, append new post, redirect home.
"""
global next_id
error = None

if request.method == "POST":
title   = request.form.get("title", "").strip()
content = request.form.get("content", "").strip()

# Basic validation
if not title or not content:
error = "Both title and content are required."
else:
new_post = {
"id":      next_id,
```

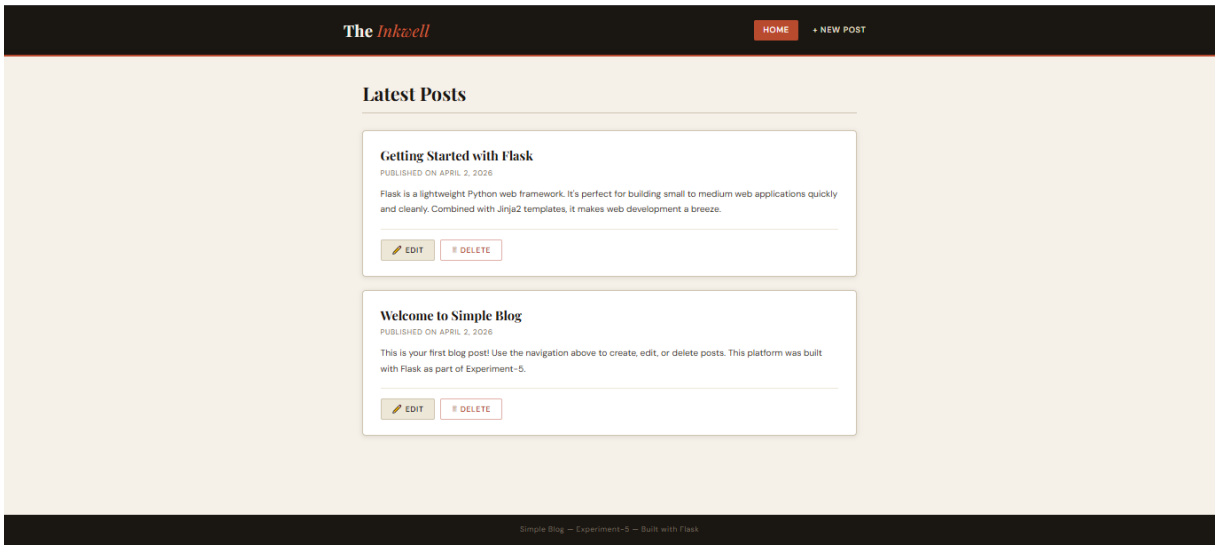
```
"title":   title,
"content": content,
"date":    datetime.now().strftime("%B %d, %Y")
}
posts.append(new_post)
next_id += 1
return redirect(url_for("index"))

return render_template("create.html", error=error)


# -------------------------------------------------------
# EDIT ROUTE — Update an existing blog post
# -------------------------------------------------------
@app.route("/edit/<int:post_id>", methods=["GET", "POST"])
def edit(post_id):
"""
GET  -> Show the edit form pre-filled with existing data.
POST -> Save updated title/content, redirect home.
"""
# Find the post by ID
post = next((p for p in posts if p["id"] == post_id), None)

if post is None:
return "Post not found.", 404

error = None
```

```
if request.method == "POST":
title   = request.form.get("title", "").strip()
content = request.form.get("content", "").strip()

if not title or not content:
error = "Both title and content are required."
else:
# Update in place
post["title"]   = title
post["content"] = content
post["date"]    = datetime.now().strftime("%B %d, %Y") + " (edited)"
return redirect(url_for("index"))

return render_template("edit.html", post=post, error=error)


# -------------------------------------------------------
# DELETE ROUTE — Remove a blog post
# -------------------------------------------------------
@app.route("/delete/<int:post_id>", methods=["POST"])
def delete(post_id):
"""Remove the post with the given ID and redirect to home."""
global posts
posts = [p for p in posts if p["id"] != post_id]
return redirect(url_for("index"))
```

```
# ----------------------------------------------------
# Run the development server
# ----------------------------------------------------
if __name__ == "__main__":
app.run(debug=True)
```

❖ Output:


The Inkwell
HOME
+ NEW POST
Create New Post
POST TITLE
Enter a compelling title...
CONTENT
Write your post here...
PUBLISH POST
CANCEL
Simple Blog — Experiment-5 — Built with Flask


The Inkwell
HOME
+ NEW POST
Latest Posts
Getting Started with Flask
PUBLISHED ON APRIL 2, 2026
Flask is a lightweight Python web framework. It's perfect for building small to medium web applications quickly and cleanly. Combined with Jinja2 templates, it makes web development a breeze.
EDIT
DELETE
Welcome to Simple Blog
PUBLISHED ON APRIL 2, 2026
This is your first blog post! Use the navigation above to create, edit, or delete posts. This platform was built with Flask as part of Experiment-5.
EDIT
DELETE
Simple Blog — Experiment-5 — Built with Flask